Divisibility Rules

A number is **divisible** by another number if it divides evenly *without a remainder.*
For example, 8 is divisible by 4 since $8 \div 4 = 2$ R0.
The following divisibility rules apply to whole numbers.

Divisor: 2

Divisibility Rule:

Any whole number that ends in 0, 2, 4, 6, or 8 is divisible by 2.
(i.e. Any even number is divisible by 2).

Example:

78558 is divisible by 2 since the last digit is 8.

Non-Example:

78559 is not divisible by since the last digit is 9.

Divisor: 3

Divisibility Rule:

If the sum of the digits is divisible by 3, then the whole number is also divisible by 3.

Example:

1539 is divisible by 3 since $1 + 5 + 3 + 9 = 18$ and 18 is divisible by 3.

Non-Example:

1540 is *not* divisible by 3 since $1 + 5 + 4 + 0 = 10$ and 10 is *not* divisible by 3.

Divisor: 4

Divisibility Rule:

If the last two digits of the whole number form a two-digit number that is divisible by 4, then the whole number is also divisible by 4.

Example:

19236 is divisible by 4 since the last two digits, 36, form a number that is divisible by 4.

Non-Example:

19237 is *not* divisible by 4 since the last two digits, 37, form a number that is *not* divisible by 4.

Divisor: 5

Divisibility Rule:

Any whole number that ends in 0 or 5 is divisible by 5.

Example:

87235 is divisible by 5 since the number ends in 5.

Non-Example:

87234 is *not* divisible by 5 since the number ends in 4.

Divisor: 6

Divisibility Rule:

If a whole number is divisible by 2 AND 3, then it is divisible by 6.

Example:

5622 is divisible by 6 since this number is divisible by 2, it ends in 2, and by 3

(5+6+2+2=15 and 15 is divisible by 3).

Non-Example:

5630 is not divisible by 6 since the number is divisible by 2, it ends in 0, but not by 3 (5+6+3+0=14 and 14 is not divisible by 3).

Divisor: 8

Divisibility Rule:

If the last three digits of the whole number form a three-digit number that is divisible by 8, then the whole number is also divisible by 8.

Example:

78160 is divisible by 8 since the last three digits, 160, form a number that is divisible by 8 (160 ÷ 8 = 20).

Non-Example:

78161 is *not* divisible by 8 since the last three digits, 161, form a number that is not divisible by 8.

Divisor: 9

Divisibility Rule:

If the sum of the digits is divisible by 9, then the whole number is also divisible by 9.

Example:

178245 is divisible by 9 since 1+7+8+2+4+5=27 and 27 is divisible by 9 (27 ÷ 9 = 3)

Non-Example:

178243 is *not* divisible by 9 since 1+7+8+2+4+3=25 and 25 is *not* divisible by 9

Divisor: 10

Divisibility Rule:

Any whole number that ends in 0 is divisible by 10.

Example:

149830 is divisible by 10 since it ends in 0.

Non-Example:

149835 is *not* divisible by 10 since it ends in 5.